

3.1: Complex Numbers

Definition. (Imaginary and Complex Numbers)

We can extend the Real numbers by defining:

$$i = \sqrt{-1}.$$

A **complex number** is a number of the form $a + bi$ where

- a is the real part of the complex number, and
- bi is the imaginary part of the complex number.

If $b = 0$, then $a + bi$ is a real number.

If $a = 0$ and $b \neq 0$, then the complex number is called an **imaginary number**.

An imaginary number is an *even* root of a negative number.

$$\begin{array}{ccc} & 5 + 2i & \\ \text{Real part} \nearrow & & \nwarrow \text{Imaginary part} \end{array}$$

Example. Express the following complex numbers in standard form:

$$\sqrt{-9}$$

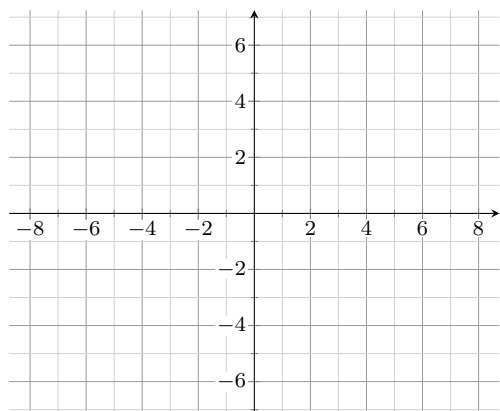
$$5 + \sqrt{-3}$$

$$4 - \sqrt{-16}$$

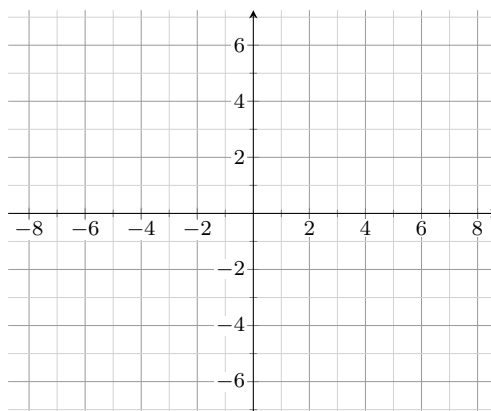
$$3 - \sqrt{25}$$

Example. Graph the following complex numbers:

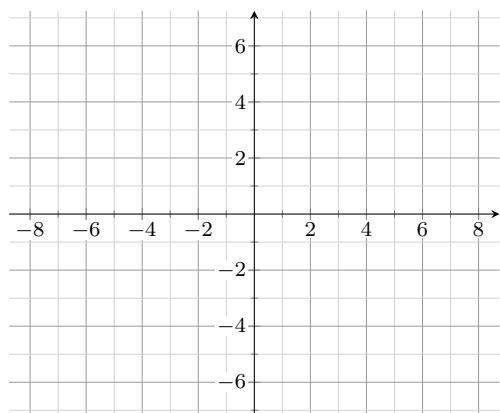
$$1 + i$$



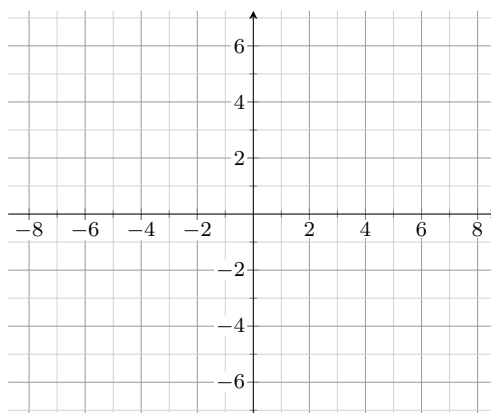
$$-2 + 3i$$



$$3 - 4i$$



$$-5 - 2i$$



Definition. (Addition and Subtraction of complex numbers)

Given complex numbers $a + bi$ and $c + di$,

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

$$(a + bi) - (c + di) = (a + c) - (b + d)i$$

Example. Evaluate the following:

$$(3 + 2i) + (1 - 4i)$$

$$(-4 - 2i) - (-1 + 5i)$$

$$(2) - (-1 + 2i)$$

$$(5i) + (3 + i)$$

Definition. (Multiplying complex numbers)

Given complex numbers $a + bi$ and $c + di$,

$$\begin{aligned}(a + bi) \cdot (c + di) &= ac + adi + bci + bdi^2 \\ &= ac + adi + bci - bd \\ &= (ac - bd) + (ad + bc)i\end{aligned}$$

Example. Evaluate the following:

$$(3 + 2i) \cdot (1 - 4i)$$

$$(-4 - 2i) \cdot (-1 + 5i)$$

$$(2) \cdot (-1 + 2i)$$

$$(5i) \cdot (3 + i)$$

Definition. (Dividing complex numbers)

Given complex numbers $a + bi$ and $c + di$, division is done via the *complex conjugate*:

$$\begin{aligned}\frac{c + di}{a + bi} \cdot \frac{a - bi}{a - bi} &= \frac{(c + di)(a - bi)}{(a + bi)(a - bi)} \\ &= \frac{ac - bci + adi - bdi^2}{a^2 - abi + abi - b^2i^2} \\ &= \frac{(ac + bd) + (ad - bc)i}{a^2 + b^2}\end{aligned}$$

Example. Evaluate the following:

$$\frac{3 + 2i}{1 - 4i}$$

$$\frac{-4 - 2i}{-1 + 5i}$$

$$\frac{2}{-1 + 2i}$$

$$\frac{3 + i}{5i}$$

Example. Given $f(x) = x^2 - 5x + 2$, evaluate $f(3 + i)$.

Example. Given $f(x) = \frac{2 + x}{x + 3}$, evaluate $f(10i)$.

Example. Find i raised to different powers: $i, i^2, i^3, \dots$, then evaluate i^{35} .